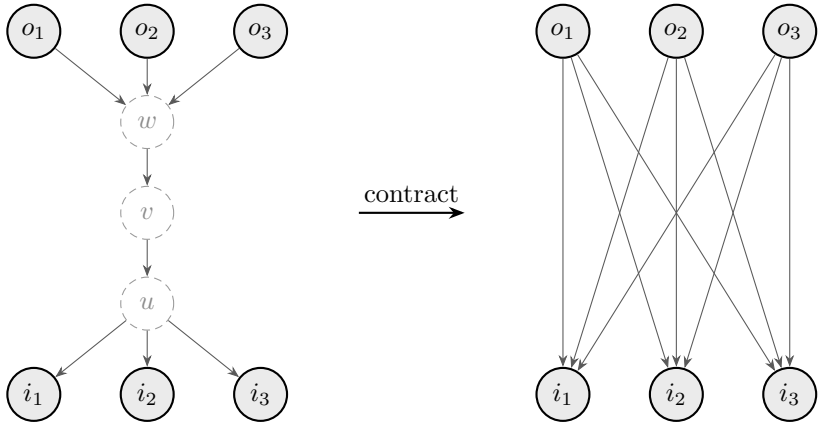


(a) uncontracted
$|E| = 8$, $o \rightsquigarrow i$ path $= 4$

(b) contracted
$|E| = 9$, $o \rightsquigarrow i$ path $= 1$

Figure 1: Contraction shortens input–output paths but can multiply edges. Edges point from a vertex to the vertices it depends on, so backward slicing reads downwards. Solid vertices are kept (inputs, outputs and `@doc`-annotated values); dashed vertices are dropped. Dropping α splices it out by rewiring α's consumers to α's frontier — its nearest kept ancestors — so in (b) every o_j points straight at $\{i_1, i_2, i_3\}$. The chain w, v, u collapses to a single hop, but its $3+2+3 = 8$ edges become $3 \times 3 = 9$. In general a chain of L dropped vertices between p inputs and s outputs replaces $p + s + L - 1$ edges with $p \cdot s$, an increase whenever $p \cdot s > p + s + L - 1$.